ModelIt!

Student Activity Pack

How Does a Neural Network Learn?

Modeling Artificial Intelligence Training, Weights, and Pattern Recognition

Grade 11L3 | NGSS HS-ETS1-1, HS-ETS1-4 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think happens inside an artificial intelligence system when it recognizes a face or generates text?**

2. **How do you think AI systems like ChatGPT or image generators actually learn — what goes in and what comes out?**

3. **Draw a diagram showing how you think a computer could be trained to distinguish between photos of cats and dogs.**

4. **What concerns do you have about AI systems making important decisions about people's lives, like medical diagnosis or hiring?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Training Data Volume | Training Data Quality | Network Depth | Learning Rate | Prediction Accuracy | Overfitting Risk | Computational Cost

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Balanced Training

Moderate data volume, high quality, moderate depth, optimal learning rate

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Scale-Up Experiment

Maximum data volume and network depth

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Biased Data Scenario

Optimal architecture but low-quality, unrepresentative training data

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Regularization Strength — Techniques like dropout or weight decay that deliberately constrain the network's ability to memorize training data — forcing it to learn generalizable patterns at the cost of slightly lower training accuracy

Inference Latency — The time required for a trained model to process a single input and produce an output — critical for real-time applications like autonomous vehicles or medical alerts where milliseconds matter

Catastrophic Forgetting — The tendency of neural networks to lose previously learned capabilities when trained on new data — a fundamental challenge for creating AI systems that continuously learn and improve over time

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a machine made of simple math operations learn to see faces, understand language, and create art — and what does it actually understand? Use evidence from your simulation to support your answers.

1 How does your model demonstrate why training data quality is arguably more important than network architecture for real-world AI applications?

2 What does the overfitting phenomenon reveal about the difference between memorization and understanding in neural networks?

3 Why does the biased data scenario have disproportionate real-world impact compared to other sources of prediction error?

4 How would your model change if training data were unlimited and perfectly representative — would that solve the understanding problem?

5 What are the limitations of modeling AI training when real-world neural networks have billions of parameters interacting in ways we cannot fully trace?

8 STEM Challenge: Design a Fair AI System for a Real-World Application

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YOUR ENGINEERING MISSION

Design a neural network-based system for a real-world application (medical diagnosis, college admissions, criminal justice, or hiring) that maximizes accuracy while ensuring fairness across all demographic groups.

THE SCENARIO

A hospital network wants to deploy an AI system that predicts which emergency room patients are most likely to have a heart attack within 24 hours, allowing faster treatment. However, early testing reveals that the system is significantly less accurate for Black patients because the training data underrepresents this population and uses historical treatment data that reflects existing healthcare disparities. Your team must redesign the training approach to ensure the system works equally well for all patients.

Engineering Design Thinking

1. WHAT SPECIFIC CHANGES TO TRAINING DATA QUALITY AND VOLUME WOULD YOU MAKE TO ADDRESS THE ACCURACY DISPARITY?

2. WHAT NETWORK ARCHITECTURE DECISIONS (DEPTH, REGULARIZATION) WOULD HELP PREVENT OVERFITTING TO THE MAJORITY POPULATION'S PATTERNS?

3. HOW WOULD YOU TEST THE SYSTEM'S PERFORMANCE SEPARATELY FOR DIFFERENT DEMOGRAPHIC GROUPS BEFORE DEPLOYMENT?

4. WHAT ONGOING MONITORING WOULD YOU REQUIRE AFTER DEPLOYMENT TO DETECT IF THE SYSTEM DEVELOPS NEW BIASES OVER TIME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Machine Learning Engineers design and train neural networks at companies like Google DeepMind, OpenAI, Anthropic, Meta AI, and Microsoft Research, earning $130,000-$300,000/year. AI Ethics Researchers investigate bias, fairness, and safety in AI systems at universities, nonprofits, and technology companies, earning $90,000-$180,000/year. Data Scientists who prepare and curate training datasets are critical to AI quality, earning $100,000-$170,000/year.